

Data Analysis Tests to Detect Red Flags of Fraud

Tests to Detect Billing Schemes

- Review changes to the vendor master file, including new vendors added and address changes.
- Extract vendors with incomplete profiles, especially those with missing telephone numbers or tax ID numbers.
- Match the vendor master file to the accounts payable register.
- Match the vendor master file to the employee master file on various key fields, such as address or tax ID number.
- Identify vendors with more than one vendor number.
- Extract multiple vendors with the same address.
- Identify vendors with PO Boxes.
- Identify vendors with significant changes in activity (increase or decrease).
- Identify duplicate purchase order numbers, credits, and invoices.
- Identify purchases without purchase orders by vendor and issuer.
- Identify credits, receipts, and invoices not in proper sequence or range.
- Review transactions just under authorization thresholds or even-dollar payments.
- Identify payments above individual authorization limits, or split payments exceeding authorization limits.
- Review payments with little or no sequence between invoice numbers.
- Review transactions by user ID for unauthorized or unusual users.
- Review transactions posted on weekends, holidays, or at other unusual times.
- Identify disbursements in which the payment approval party is the same as purchase approval party.
- Perform a trend analysis of vendor payments.
- Identify duplicate vendor numbers on vendor master file.
- Sample vendor open invoices for confirmation with vendor.
- Report on variances between purchase orders, delivery documents, and invoices.
- Compare expenses to budgeted or prior-period amounts.
- Identify significant increases in average unit prices of goods.
- Summarize card use by employee and sort high to low.
- Identify payments/card transactions to vendors normally associated with personal purchases.
- Extract multiple charges of the same product type below a predefined card expense limit.
- Extract all delivery addresses that do not correspond to a company location.

Tests to Detect Check Tampering Schemes

- Test for out-of-range checks.
- Identify checks issued to vendors with names that are similar to those of known vendors.
- Extract all payments equal to zero.
- Extract all reconciling items per the bank reconciliation and summarize for reasonableness.

- Extract users who can write checks or initiate electronic payments and post entries to the general ledger.
- Identify checks with no related system invoices.
- Identify payments charged to dormant or unused accounts.
- Extract manual checks and summarize by payee and issuer.
- Review transactions for round-dollar payments.
- Identify duplicate purchase order numbers, credits, and invoices.
- Summarize payment activity by general ledger account and review for reasonableness.
- Identify purchases without purchase orders by vendor and issuer.
- Review transactions just under authorization thresholds or even-dollar payments.
- Identify payments above individual authorization limits, or split payments exceeding authorization limits.
- Review transactions by user ID for unauthorized or unusual users.
- Review transactions posted on weekends, holidays, or at other unusual times.

Tests to Detect Payroll Schemes

Ghost Employee Schemes

- Identify persons on the payroll with no employment history and no performance evaluations.
- Identify persons on the payroll with no time off for vacations or sick leave.
- Review trends in payroll amounts by general ledger account.
- Extract all round-dollar payments.
- Identify duplicate employee names, addresses, or phone numbers.
- Identify invalid Social Security numbers.
- Compare hiring and termination dates in the employee master file to pay-period dates.
- Compare pay-period dates with employee start and termination dates.
- Extract users who can issue paychecks and can add new employees in the payroll and time card systems.
- Extract employees with multiple address changes in the last year.
- Extract multiple payroll deposits to the same bank account in a single pay period.
- Identify employee addresses that are PO Boxes.
- Identify employees with no pay increases or more than one pay increase in the past year.
- Stratify manual checks to review for individuals who were issued more than one manual check.

False Salaries and Wages Schemes

- Stratify on hourly rates, hours worked, net pay amount, or overtime hours.
- Compare salaried employee gross pay from one pay period to the next.
- Compare salaried employee net pay from one pay period to the next.
- Compare pay rate with ranges for employee classification.
- Identify changes in salary amounts.

- Calculate the percentage of bonus pay to gross pay (on an employee-by-employee basis) and sort from high to low.
- Compare bonus payments to budgeted or prior-year amounts on an employee-by-employee basis.
- Compare hours reported per time card system to payroll system.
- Compare overtime hours to budgeted or prior-year amounts by department.
- Compare pay rates in the payroll system to those in the employee master file.
- Extract all employees with over X hours per pay period.
- Recalculate gross pay.
- Recalculate net pay.
- Calculate the average payroll per employee and sort from high to low.
- Identify changes in exemptions, gross pay, net pay, hourly rates, and salary amounts.
- Review overtime pay, special pay, reimbursements added to paychecks, and bonuses.
- Compare time card rates and pay rates to payroll register and report variances.
- Extract all payroll checks in which gross pay exceeds set amount.
- Extract employees with pay increases greater than X percent.

Commissions Schemes

- Age open invoices by salesperson.
- Identify sales to customer accounts that had been dormant for more than X months.
- Identify new customer accounts added during the review period.
- Extract customers without a tax ID number (or with other missing information).
- Stratify on commission amount.
- Extract customer accounts receivable balances that exceed customer credit limits.
- Compare the customer master file to the sales register.
- Extract any salespersons who can create customer accounts and enter sales orders or adjustments.
- Perform a trend analysis of commission-to-sale ratios by salesperson.
- Sequence possible duplicate sales invoices based on the absolute value of the invoice and customer.
- Calculate the ratio of the largest sale to next largest sale by customer.
- Extract customer sales that exceed the X-month average sales from that customer by a specified percentage.
- Summarize commissions paid by product line, region, and salesperson.
- Review bonuses and commissions.
- Recalculate commissions based on current year sales and other required performance measures.
- Classify sales prices by salesperson and calculate an average price per salesperson.

Tests to Detect Expense Reimbursement Schemes

- Summarize expense reimbursements by employee.
- Compare expense reimbursement levels to budgeted and prior-year amounts.
- Compare dates and times of expenses to employees' work schedules.
- Age employee payments by check date.

- Stratify by expense reimbursement amount.
- Identify reimbursement claims with missing descriptions such as dates, times, or purposes of expenses.
- Identify potential duplicate expenses based on absolute value and receipt date of expense.
- Extract all reimbursements in round-dollar amounts.
- Extract all reimbursements just below approval limit.
- Identify employees whose expenses differ significantly from the prior year.
- Identify employees whose expenses are out of line with those of other employees in similar positions.
- Match expense claim amounts to amounts on corporate credit card.
- Identify claims for use of personal vehicle and rental car for same period.
- Identify overlapping travel claims.

Tests to Detect Theft of Cash Receipts Schemes

- Extract customers with credit balances.
- Identify higher-than-average uncollectable accounts.
- Compare remittances to open receivables and report variances.
- Generate invoice summaries by customer or invoice amount.
- Identify credits taken beyond discount terms of payment dates.
- Identify adjustments to discounts.
- Extract sales with more than X percent discount and summarize by employee.
- Identify accounts with even balances.
- Identify gaps in the sequencing of invoices generated.
- Identify duplicate invoices, credits, or receipts.
- Identify credits, receipts, and invoices that are not in proper sequence or range.
- Identify variances between delivery documents and invoices.
- Calculate days to payment by client, by sales representative, or by region, and calculate carrying charges.
- Identify accounts with oldest activity for sales follow-up.
- Extract invoices with partial payments.
- Extract customers with incomplete profiles, especially those missing a telephone or tax ID number.
- Identify customers added during the period under review.
- Match customer master file to employee master file on various key fields.
- Compare days in collection by customer.
- Extract variances in matching remittances to open receivables.
- Identify high value credit notes, balances, and invoices—by customer and by clerk.
- Age total receivables.
- Summarize net sales by employee and extract top X employees with low sales.
- Summarize by location—discounts, returns, inventory adjustments, accounts receivable write-offs, and voids charged.

- Summarize by employee—discounts, returns, inventory adjustments, accounts receivable write-offs, and voids charged.
- List top X employees by dollar size of discounts, refunds, inventory adjustments, accounts receivable write-offs, and sale voids.
- Compare adjustments to inventory to void/refund transactions summarized by employee.
- Summarize user access for sales, accounts receivable, inventory, and general ledger systems for segregation of duties reviews.
- Summarize user access for sales, accounts receivable, inventory, and general ledger systems by non-business hours.
- Calculate the average days from delivery of goods or services to billing date and to receipt of payment by clerk and by customer.
- Identify credits/debits to previously unused accounts.
- Summarize by employee the difference between the cash receipt report and the sales register system.

Tests to Detect Inventory Schemes

- Identify unusual delivery addresses.
- Stratify inventory adjustments by employee, extended cost, location, etc.
- Identify negative receipt quantities.
- Calculate stock inventory turnover by item.
- Identify duplicate items or serial numbers.
- Identify duplicate inventory listing by amount and description, as well as quantity and amount.
- Extract all inventory coded as obsolete that possesses reorder points within the inventory system.
- Identify obsolete inventory by sorted turnover analysis.
- Identify receipts per receiving reports in the receiving system that do not agree to the receipts per the accounts payable invoices.
- Identify inventory receipts per inventory item that exceed the economic order quantity or maximum for that item.
- Identify items below the standard margins.
- Extract inventory with a negative or zero price.
- Calculate the ratio of the largest sale to the next largest sale by customer.
- Identify items with yearly volume over/under total quantity ordered.
- Determine percentage change in sales, price, or cost levels by product/vendor.
- Calculate the difference between actual and standard costs.
- Isolate stock where cost is greater than retail price.
- Match stock receipts with vendor ledger and report variances.
- Calculate returns and shortages by vendor and by clerk.
- Test for duplicate parts or descriptions.
- Calculate number and value of write-offs and strike-offs by location.
- Calculate number and value of items with disposal followed by reorder transactions.

- Identify items with high return or allowance rates.
- Calculate number and value of items sent to scrap by clerk.
- Calculate shortages and overages by ordering and receiving clerk and by vendor.

Tests to Detect Corruption Schemes

- Identify duplicate payments.
- Stratify vendor payments by approval limits, especially those directly below the approval limit.
- Calculate the ratio of the largest purchase to the next largest purchase by vendor.
- Summarize by vendor the number of inferior goods based on number of returns.
- Match the vendor master file to the employee master file on various key fields.
- Extract round-dollar payments and summarize by vendor.
- Identify payments with little or no sequence between invoice numbers.
- Summarize invoice payments by type of purchase and identify areas with fewer than three vendors.
- Calculate total purchases by ordering clerk and by vendor.
- Compute total business volume by agent and by vendor.
- Match contract price to invoice and report on variances.
- Stratify inventory actual price to standard price.
- Identify higher-than-average inventory costs.
- Compare order quantity to optimal reorder quantity.
- Identify delivery of inventory to employee address by joining employee address to shipment address file.
- Identify delivery of inventory to address not designated as a business address.
- Compare inventory levels and turnover rates.
- Match quantities ordered to quantities received.
- Calculate unit price by product.
- Identify cases in which inventory was written off and then a new purchase was made by comparing total write-off and purchase quantities by stock number.
- Calculate the total number and total value of contracts by contracting officer and by vendor.
- Calculate average value of contracts awarded by vendor.
- Calculate the percentage contract awards by vendor (contracts won divided by total number of bids).
- Calculate the percentage of sole-sourced contracts by vendor (contracts sole-sourced divided by total number of contracts).

Tests to Detect Financial Statement Fraud Schemes

- Track year-to-date activity for large operating accounts.
- Compare summaries by major account in any order (i.e., low-to-high, high-to-low).
- Identify employees who made a large number of adjustments or reversing entries.
- Calculate and sort percentage variances in accounts between periods.
- Re-compute trial balances.
- Prepare account reconciliations.

- Identify large adjustment transactions.
- Review transactions posted at unusual times.
- Review transactions posted on unusual dates.
- Review suspense transactions activity for appropriate and timely clearing and offsets.
- Review transactions split among a large number of general ledger accounts.
- Review transactions with large even-dollar amounts.
- Review debit balances in accounts that have normal credit balances, and vice versa.
- Review manual postings to auto-posted accounts.